Bug #15 — WDM phitilde betainc NaN at boundary

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SYMPTOM
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Initial log_like = NaN in the WDM gb_and_foreground smoke run.
Probing showed the WDM data array was 100% NaN, while the TD/FD
inputs were clean.

ISOLATION
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Reduced to debug_wdm_transform_nan.py operating on a saved
td_sig.npy (shape (3, 1555200), no NaN, max |a| ~ 5.27e-20):

  WDM window: shape (2160,), nan = 2 at indices [270, 1890]
              omega at NaN = +/- 3.272492e-3 = +/- (A + B) = +/- pi/960
  WDM signal (post-transform): nan = 100% across all channels
  Lookup table (table_cos, table_sin): 12000 / 12000 NaN
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Root cause

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LOCATION
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WDMSettings.phitilde in src/lisatools/domains.py around L1528:

    A = self.A
    B = dOmega - 2 * A
    z = self.xp.zeros(omega.shape[0])
    beta_inc_calc = (np.abs(omega) >= A) & (np.abs(omega) <= A + B)
    x = (np.abs(omega[beta_inc_calc]) - A) / B
    y = special.betainc(WAVELET_FILTER_CONSTANT,
                        WAVELET_FILTER_CONSTANT, x)
    z[beta_inc_calc] = insDOM * np.cos(y * np.pi / 2.0)

WHY IT FAILS
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At omega = +/- (A + B), the mathematical value of x is exactly 1.0,
but float arithmetic gives 1.0 + 1 ULP = 1.0000000000000002.
scipy.special.betainc(4, 4, x) is defined only on x in [0, 1] and
returns NaN for any x > 1.0:

    betainc(4, 4, 1.0)               = 1.0
    betainc(4, 4, 1.0000000000000002) = nan

PROPAGATION
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1. Two NaN entries land in WDMSettings.window.
2. In FDSignal.wdmtransform the FD-then-IFFT path multiplies
   before_ifft *= base_window[None, None, :], seeding 2 * (Nf + 1)
   = 1442 NaN cells per channel.
3. np.fft.ifft along the last axis mixes all bins, spreading NaN
   over the entire (nchannels, Nf + 1, Nt) array (100% NaN).
4. The same path built the lookup table, so table_cos and table_sin
   were 100% NaN as well.
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Fix and follow-up

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FIX (one-line, minimal)
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Clip x into [0, 1] before betainc. The mathematical boundary
value at x = 1 is z = insDOM * cos(pi / 2) = 0, so clipping is
exact at the boundary and only suppresses the ULP overshoot:

    x = np.clip((np.abs(omega[beta_inc_calc]) - A) / B, 0.0, 1.0)
    y = special.betainc(WAVELET_FILTER_CONSTANT,
                        WAVELET_FILTER_CONSTANT, x)

POST-FIX VERIFICATION (debug_wdm_transform_nan.py)
--------------------------------------------------
  WDM window:                nan = 0
  before_ifft (post-window): nan = 0
  after_ifft:                nan = 0
  wdm.arr (active slice):    nan = 0, max |a| ~ 9e-20

FOLLOW-UP
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Rebuild wdm_lookup_n_ref_NF720_NT2160_3mo.h5 (build script
build_wdm_lookup_3mo.py). The cached table on disk was poisoned
by the same bug and must be regenerated for the fix to flow
through GB likelihood and swap_ll kernels.
```